
DSC 140A - Homework 05

Homework 05 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using L^AT_EX with our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, convert them into PDF, and then merge the PDF with your math problems solutions generated in L^AT_EX.

Problem 1.

In this problem, you will derive the solution to the ridge regression optimization problem.

Recall that the ridge regression regularized risk function is

$$\tilde{R}(\vec{w}) = \frac{1}{n} \sum_{i=1}^n (\vec{\phi}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 + \lambda \|\vec{w}\|^2.$$

Here, $\vec{\phi}(\vec{x})$ is a feature map. Also recall that this risk function can be equivalently written in matrix-vector form as

$$\tilde{R}(\vec{w}) = \frac{1}{n} \|\Phi \vec{w} - \vec{y}\|^2 + \lambda \|\vec{w}\|^2,$$

where Φ is the design matrix; its i th row is $\vec{\phi}(\vec{x}^{(i)})$.

- a) Show that $\tilde{R}(\vec{w}) = \frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - 2\vec{w}^T \Phi^T \vec{y} + \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w}$.
- b) So far this quarter, we have seen a few vector calculus identities. For example, we know that $\frac{d}{d\vec{w}} (\vec{w}^T \vec{w}) = 2\vec{w}$.
Using these identities, show that

$$\frac{d}{d\vec{w}} \tilde{R}(\vec{w}) = \frac{1}{n} (2\Phi^T \Phi \vec{w} - 2\Phi^T \vec{y}) + 2\lambda \vec{w}.$$

- c) Show that the minimizer of $\tilde{R}(\vec{w})$ is $\vec{w}^* = (\Phi^T \Phi + n\lambda I)^{-1} \Phi^T \vec{y}$.

Problem 2.

Suppose you are modeling the relationship between a single input x and a target y . You suspect the true relationship is quadratic, so you decide to use the feature map $\vec{\phi}(x) = (x, x^2)^T$. You have the following training data:

$x^{(i)}$	y_i
0	1
1	2
2	5

- a) Compute $\vec{\phi}(x^{(i)})$ for each training point and construct the augmented design matrix Φ .
- b) The least squares solution in feature space is $\vec{w}^* = (1, 0, 1)^T$. Write the prediction function $H(x)$ explicitly as a function of x alone — not in terms of ϕ . Is this function linear in x ?

- c) A new point arrives with $x = 3$. Using the two-step prediction procedure from lecture, compute $H(3)$.
- d) A classmate argues: “Since we train a linear model in feature space, the prediction function must also appear linear when plotted against the original input x .” Is this correct? Explain in one to two sentences.

Problem 3.

In this problem, you will explore the geometry of the Hard-SVM and how scaling and feasibility affect the choice of classifier. Recall that the Hard-SVM optimization problem is

$$\vec{w}^* = \arg \min_{\vec{w}} \|\vec{w}\|^2 \quad \text{subject to} \quad y_i (\vec{w} \cdot \vec{x}^{(i)}) \geq 1 \quad \text{for all } i,$$

and that the margin width of a classifier is $\frac{2}{\|\vec{w}\|}$.

Consider the following training dataset:

i	x_1	x_2	y_i
1	2	-1	+1
2	1	0	+1
3	-1	0	-1
4	-2	3	-1

Assume $b = 0$ throughout. Two candidate weight vectors are proposed:

- $\vec{w}_A = (1, 0)$, so $H_A(\vec{x}) = x_1$
 - $\vec{w}_B = (2, 1)$, so $H_B(\vec{x}) = 2x_1 + x_2$
- a) For each candidate, check whether every training point satisfies the Hard-SVM constraint $y_i (\vec{w} \cdot \vec{x}^{(i)}) \geq 1$ for all i . Show your work for each point under each classifier.
- b) Both classifiers satisfy the Hard-SVM constraints. Compute the margin width for each. Which would the Hard-SVM select, and why?
- c) Identify all support vectors under $\vec{w}_A$. Recall that a support vector is a training point $\vec{x}^{(i)}$ such that $y_i \vec{w} \cdot \vec{x}^{(i)} = 1$.
- d) A classmate proposes using $\vec{w}_C = (2, 0)$ instead of $\vec{w}_A$, arguing it is just a scaled version and should perform the same.
- (i) Verify that $\vec{w}_C$ satisfies all Hard-SVM constraints.
 - (ii) Compute the margin width under $\vec{w}_C$.
 - (iii) Is your classmate correct? What does this tell you about scaling $\vec{w}$ in the context of Hard-SVM?
- e) A new training point $\vec{x}^{(5)} = (0.4, 1)$ with label $y_5 = +1$ is added to the dataset.
- (i) Show that this point violates the Hard-SVM constraints under $\vec{w}_A$.
 - (ii) Explain briefly why the Hard-SVM optimization problem no longer makes sense with this new point in the dataset.
 - (iii) If we instead use a Soft-SVM, what is the minimum slack $\xi_5 \geq 0$ required so that the constraint $y_5(\vec{w}_A \cdot \vec{x}^{(5)}) \geq 1 - \xi_5$ is satisfied?

Problem 4.

In this problem, you will explore how the regularization parameter C controls the tradeoff between margin width and classification accuracy in the Soft-SVM. Recall the hinge loss:

$$\ell_{\text{hinge}}(H(\vec{x}), y) = \max\{0, 1 - y H(\vec{x})\},$$

and the regularized SVM risk:

$$R_{\text{svm}}(\vec{w}) = \|\vec{w}\|^2 + \frac{C}{n} \sum_{i=1}^n \max\{0, 1 - y_i \vec{w} \cdot \vec{x}^{(i)}\}.$$

Consider the following training dataset with $n = 4$ points:

i	x_1	x_2	y_i
1	3.0	2	+1
2	0.8	-1	+1
3	-3.0	1	-1
4	-0.8	2	-1

Two candidate predictors (with $b = 0$) are proposed:

- $\vec{w}_A = (1, 0)$, so $H_A(\vec{x}) = x_1$
 - $\vec{w}_B = (0.5, 0)$, so $H_B(\vec{x}) = 0.5 x_1$
- a) For each predictor, compute $y_i H(\vec{x}^{(i)})$ and the hinge loss ℓ_{hinge} for every training point. Organize your answers in a table with columns: i , $y_i H_A(\vec{x}^{(i)})$, ℓ_{hinge} under $\vec{w}_A$, $y_i H_B(\vec{x}^{(i)})$, ℓ_{hinge} under $\vec{w}_B$.
 - b) Compute $R_{\text{svm}}(\vec{w}_A)$ and $R_{\text{svm}}(\vec{w}_B)$ at $C = 2$. Which predictor does the Soft-SVM prefer at this value of C ?
 - c) Now compute $R_{\text{svm}}(\vec{w}_A)$ and $R_{\text{svm}}(\vec{w}_B)$ at $C = 10$. Which predictor does the Soft-SVM prefer now?
 - d) Find the exact value of C at which both predictors have equal regularized risk. *Hint: express R_{svm} as a linear function of C for each predictor and set them equal.*
 - e) In one or two sentences, explain intuitively why the preferred predictor changes as C increases. Your answer should reference what $\|\vec{w}\|^2$ and the hinge loss terms each represent in the optimization.
 - f) As $C \rightarrow \infty$, what does the Soft-SVM approach? What condition on the data is required for this limit to be well-defined, and does this dataset satisfy it? Justify your answer.